Core pipeline

Build the snapshot index and the single processing pipeline that both events and CSV rows use.

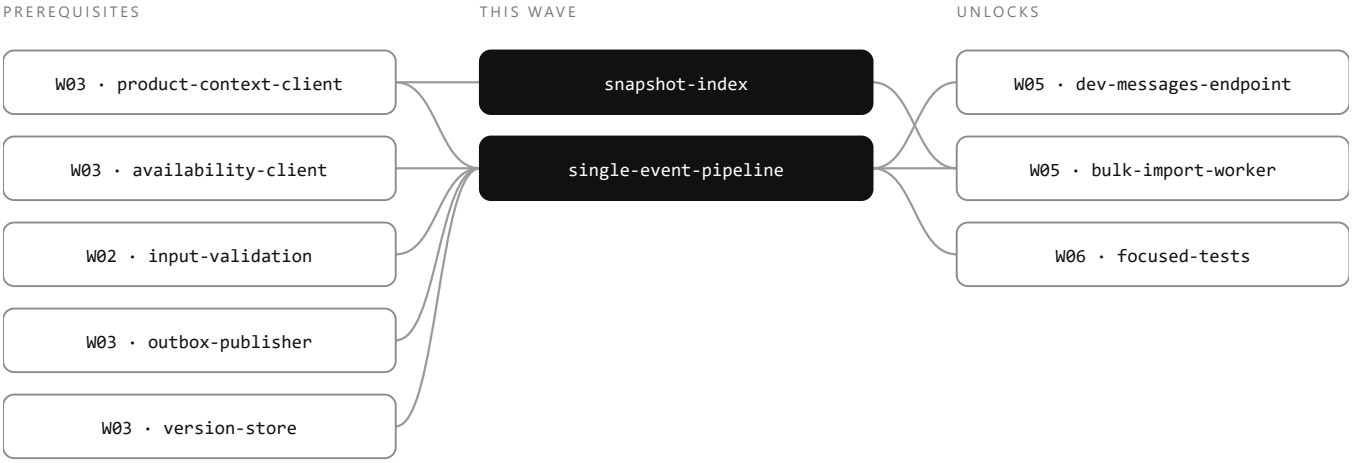
2 tasks | Requires Wave 02, 03 | Unlocks Wave 05, 06

Why this wave exists

The pipeline is the core business rule. It is the only place where validation, the version rule, enrichment, the gateway write and the outbox commit meet.

How it fits

The diagram shows this wave's tasks, the earlier-wave tasks they depend on, and the later-wave tasks that depend on them.



Hand-off. Wave 5 wires the pipeline into the dev endpoint and into the bulk worker. Both must call the same function.

Tasks

Implementing snapshot fetch and index

`snapshot-index`

Depends on: product-context-client

Fetch GET /v1/product-contexts/snapshot?countryCode=PT once per country and build an in-memory Map keyed by productId|sizeId|countryCode (about 50,000 entries). Concurrent callers share the same in-flight promise. Lookups are O(1). The index is never rebuilt per row.

`src/index/snapshotIndex.ts`

- [ ] The snapshot is requested exactly once per country, even under concurrent calls (checked with a request counter in a test).
- [ ] A key missing from the snapshot is a terminal not-found result.

Composing shared processing pipeline

`single-event-pipeline`

Depends on: availability-client, input-validation, outbox-publisher, product-context-client, version-store

Compose the single flow used by events and CSV rows. Steps: validate; quick version check (stale becomes SKIPPED_STALE); enrich from the snapshot index or the client (missing, inactive or disabled becomes REJECTED); acquire the per-stock-key lease in stock_leases; re-check the version; PUT to the gateway; then, in one transaction, commit the version and enqueue the outbox event; release the lease in a finally block. A failure after the PUT is safe to retry because the PUT is absolute.

`src/pipeline/processStockChange.ts · src/pipeline/keyLease.ts`

- [ ] The result is one of APPLIED, SKIPPED_STALE, REJECTED (with reason) or FAILED_RETRYABLE (with reason).
- [ ] Two changes for the same key run one at a time. The lease is per key and stored in the database, so it works across processes that share the database file.
- [ ] An expired lease left by a crashed process is taken over after LEASE_TTL_MS.
- [ ] The gateway write happens before the version commit.

Risks and decisions

- Order matters: gateway write first, then the version commit together with the outbox row.
- The lease TTL (LEASE_TTL_MS, default 30000) must exceed the longest gateway call including retries. Record this in docs/decisions.md.

Verification

```
npm run typecheck
npm test -- pipeline snapshot
```

Done when

1. Every acceptance box is checked and the verification commands pass.
2. The session todos for this wave are marked done.
3. The final message includes a hand-off note for the next wave.

Agent brief for this wave: AGENT.md in the same folder.